



The Slip Outlasted the App

Building and Piloting a Paper-Till, Sticker-Tag Canteen Pre-Order System, and How Its Order Log Became an Informal Lateness Flag

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Abstract. Primary-school canteens that pre-order via a commercial app carry ongoing licence costs, exclude families without a compatible device, and generate order data the canteen itself cannot inspect. We report the design, staggered four-week rollout, and eight-week pilot of a paper-till, sticker-tag system built to replace such an app at a 382-student primary school: a numbered order slip and matching peel-off sticker tag, a volunteer-staffed till, and a carbon-duplicate log time-stamped at the point of sale. Across 24 canteen days and 6,140 processed orders, slip-sticker mismatch requiring volunteer correction fell from 8.7% in week one to 1.9% by week eight, while per-order processing time rose modestly over the retired app's logged baseline, a trade-off the canteen judged acceptable against the app's ongoing costs. An ablation comparing carbon-duplicate against single-ply order-slip stock left the mismatch rate statistically unchanged (n.s.); we retain the duplicate for the audit trail it leaves rather than for any gain in accuracy. A document-analysis pass over volunteer handover notes and front-office correspondence surfaced an unplanned second use: staff began cross-referencing a student's order timestamp against the morning cutoff to corroborate, and in a few cases first raise, a suspected late arrival. We report this use descriptively, neither endorsing nor formally evaluating it, and discuss what a canteen log gains, and risks, once it is read as evidence of something other than lunch.

Introduction

Commercial pre-order apps have become a familiar fixture of primary-school canteens, replacing the exercise-book order lists and coin tins that preceded them. They are also a recurring source of friction: a per-transaction fee passed on to families or absorbed by the canteen, a smartphone or card the ordering parent must hold, and an order log the canteen itself cannot query without contacting the vendor. Canteen research has traditionally centred on nutrition and food-purchasing outcomes rather than the ordering infrastructure itself [1], leaving the everyday mechanics of how an order actually gets from a family to a till comparatively undescribed. When a parent-run canteen retired its commercial ordering app at the start of a school term, it did not replace it with another digital system but with a paper one: a numbered order

slip completed at drop-off, a matching peel-off sticker affixed to the slip's carbon duplicate, and a volunteer-operated till that time-stamps each transaction by hand.

This paper reports the system's design and its eight-week pilot deployment, treating the build itself as the contribution rather than a hypothesis to be tested against it. We describe the slip-and-sticker mechanics, the staggered by-year-level rollout, and the volunteer training the cutover required, then report uptake, mismatch, and processing-time data across the pilot's 24 canteen days. We also document, as a secondary and independent observation, an unplanned use the system's paper log acquired midway through the pilot: front-office staff began treating a late order timestamp as an informal corroborating signal for a student's late

arrival, a use the canteen neither designed for nor was consulted on before it began.

We make three contributions, each hedged appropriately:

- We report the design and staggered rollout of a paper-till, sticker-tag pre-order system built to replace a commercial canteen app, including the volunteer procedures the cutover required.
- We report descriptive uptake, mismatch, and processing-time measures across an eight-week pilot, finding mismatch requiring volunteer correction fell substantially over the pilot while processing time rose modestly relative to the retired app's logged baseline.
- We document, via a document-analysis pass over volunteer handover notes and front-office correspondence, an emergent informal use of the system's order-timestamp log as a lateness-corroboration signal, reporting the use descriptively rather than evaluating its accuracy.

The remainder of the paper reviews related work on canteen ordering technology, workaround behaviour in information systems, and function creep in administrative data; describes the system and pilot method; reports the eight-week deployment; and discusses what a paper log invites once it is read for something other than its stated purpose.

Related work

Canteen and cafeteria ordering technology research has largely focused on adoption and behaviour-change outcomes: cashless systems are reported to support point-of-sale incentive schemes such as nutrition-linked rewards [2]. Even where digital ordering does not feature, canteen operations remain a site of negotiated, unevenly enforced institutional rules, with regulatory authorities, canteen owners, and parents reporting divergent accounts of what compliance with national canteen standards actually requires in practice [3]. Attendance and administrative-data research has separately shown that school-collected records systematically diverge from self-report, shaped as much by collection procedure as by the underlying behaviour they are meant to track [4] — a caution that extends naturally to any log an institution did not build for measurement in the first place.

The information-systems literature on workarounds documents how staff routinely bend or bypass a system's intended use to close a gap between its design and their actual task, in both clinical [5], [6], [7], [8] and general organisational settings [9], [10], with recent work reframing such workarounds as informative signal for system design rather than mere deviation [11]. Function creep, the gradual use of a data-collection system beyond its original purpose, is a long-studied phenomenon in surveillance and privacy scholarship [12], recently extended to the school setting itself, where monitoring software licensed for safety purposes is shown to be readily repurposed for broader behavioural oversight [13]. Sensing and logging infrastructure built for one operational purpose is similarly shown to find a second life once its data proves useful elsewhere [14]. Prior work at this institution has traced comparable second lives for an aggregate engagement count [15] and for a discussion-log instrument scored against an independent read of the same event [16]; we extend this line to a paper log built for a single transactional purpose.

Method

The canteen, a parent-run canteen at a 382-student primary school, retired its commercial ordering app at the start of the school term, citing the app's per-transaction fee and its exclusion of families without a compatible device or card. The replacement system comprises three physical parts: a triplicate order slip (name, class, item codes, a pre-printed six-digit order number), a peel-off sticker bearing the same six-digit number in large print, and a carbon duplicate retained by the till. A parent or student completes the slip at morning drop-off and posts it, with payment, into a labelled box outside the canteen; a rostered volunteer opens the box before first bell, enters each slip's items and total into the till, affixes the sticker to the child's lunch bag at pack-up, and time-stamps the carbon duplicate with a hand ink-stamp before filing it chronologically in a ring-bound logbook.

The rollout was staggered by year level over four weeks — Kindergarten to Year 2 in week one, Years 3 and 4 in week two, and the remaining year levels folded in by week four — with a fifteen-minute volunteer briefing preceding each cohort's cutover. We logged every transaction across the eight-week pilot that followed

full cutover (24 canteen days, three per week), recording, for each order: the completion time stamped on the carbon duplicate, whether the volunteer flagged a slip-sticker mismatch requiring correction before pack-up, and the wall-clock interval between a slip's arrival in the box and its till entry.

Mismatch rate for canteen day d is the proportion of that day's orders flagged for correction:

$$M_d = \frac{N_{\text{flagged},d}}{N_{\text{total},d}} \times 100\%$$

reported per week as the mean of that week's three canteen days. We compared this study's processing-time data against the retired app's own exported transaction log for the eight weeks preceding cutover as a baseline, restricting the comparison to orders of equivalent item count to hold order complexity constant. From week five, order slips printed on carbon-duplicate stock were alternated day-by-day with single-ply stock lacking a duplicate, to test whether the duplicate itself affected mismatch rate rather than merely the audit trail it left behind. From week three, the front office's own handover notebook and internal email correspondence referencing the canteen log were retained, with the canteen coordinator's permission, and coded thematically by two of the authors for any reference to the order log's use outside its ordering purpose, resolving disagreement by discussion.

Results

Across the eight-week pilot (24 canteen days), the canteen processed 6,140 orders, a mean of 256 orders per canteen day. Correctly completed orders — slip and sticker matching without volunteer correction — rose steadily across the pilot (Figure 1), from 58.4% of week-one orders to 93.1% by week eight, with the steepest gain following the Year 5/6 cutover in week four, the cohort least used to a paper order form.

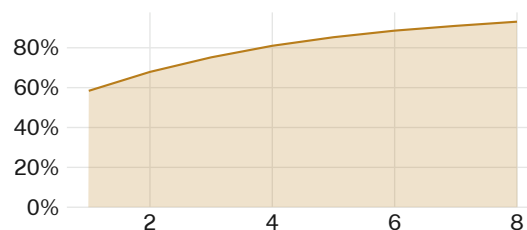


Figure 1: Weekly share of orders completed without a volunteer-flagged mismatch, rising from 58.4% in week one to 93.1% by week eight as families and student order-takers adjusted to the slip-and-sticker format.

Mismatch rate fell over the same period from a week-one mean of 8.7% to 1.9% by week eight (Figure 2), tracking the uptake curve's inverse and levelling from week six onward, after which no week's mismatch rate exceeded 2.4%.

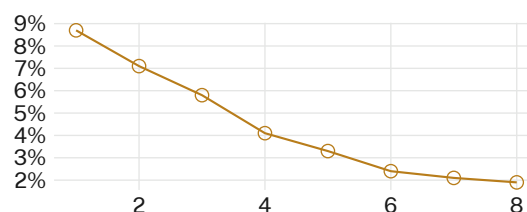


Figure 2: Weekly mean mismatch rate requiring volunteer correction, falling from 8.7% in week one to 1.9% by week eight and levelling from week six.

Per-order processing time rose modestly against the retired app's logged baseline (Figure 3): a mean of 14.2 seconds for single-item orders under the app against 19.6 seconds under the paper till, and 22.1 against 28.7 seconds for multi-item orders. The gap held roughly constant across both order-complexity categories, consistent with the added step — matching, stamping, and filing the carbon duplicate — costing a broadly fixed increment regardless of order size.

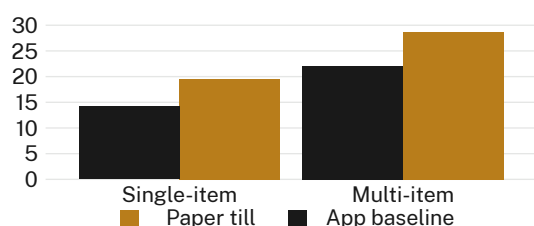


Figure 3: Mean per-order processing time, app baseline against paper-till pilot, by order complexity; the paper till adds a broadly fixed increment to each order regardless of item count.

An ablation alternating carbon-duplicate and single-ply order-slip stock daily from week five left the mismatch rate statistically indistinguishable between stock types (Figure 4): a median of 2.6% on carbon-duplicate days against 2.8% on single-ply days (n.s.), evidence that the duplicate’s contribution is to the log’s audit trail rather than to the transaction’s accuracy. We retain the carbon duplicate for the record it leaves, not for any measurable gain in mismatch rate.

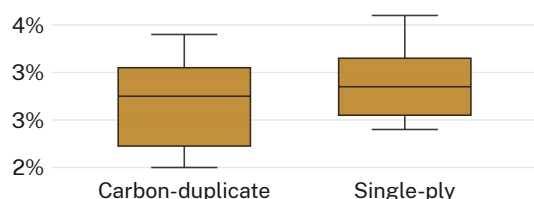


Figure 4: Daily mismatch rate under carbon-duplicate against single-ply order-slip stock from week five onward; the two distributions are statistically indistinguishable.

The document-analysis pass over front-office correspondence and the volunteer handover notebook, retained from week three, surfaced eleven instances across the pilot’s remaining six weeks in which a staff member referenced a student’s canteen order timestamp in relation to a late arrival: seven corroborating an already-flagged lateness against the morning sign-in register, three raising a lateness concern in the first instance from the timestamp alone, and one disputing a parent’s account of an on-time arrival. None of these instances followed a formal request to the canteen; the coordinator’s notes describe the practice as something front-office staff “started doing on their own” once they noticed the logbook sat, time-stamped, a few metres from the sign-in desk.

Discussion

Read together, the uptake and mismatch trends describe a system that became more reliable as its users learned it, at a modest and roughly fixed cost in processing time per order — a trade-off the canteen judged acceptable against a subscription fee and a device requirement the paper system removes entirely. The ablation’s null result suggests that trade-off buys little from the duplicate stock itself; what the duplicate buys is a retained, time-stamped, and physically inspectable record, which is precisely the property the front office found a use for. We did not design the order log to support a lateness judgement, and did not learn of the practice until the document-analysis pass was already underway; we report it descriptively rather than as a validated instrument, and take no position on whether an order timestamp is a sound basis for that judgement in any individual case. The canteen’s preference for a log it can hold and read itself, rather than depend on a vendor-hosted dashboard, follows a broader institutional pattern in which whatever enters the written record close at hand tends to be read, sooner or later, as evidence of something beyond what it was written down to track [17]. A system built to log lunch orders did not, on that account, need to try very hard to become read as something else.

Limitations

This pilot covers one canteen at one school over eight weeks, and the processing-time comparison rests on the retired app’s own exported log rather than an independently timed baseline, though the app’s export format is standard across the vendor’s other school deployments. The document-analysis pass began in week three, so any informal use of the order log in the pilot’s first fortnight is not captured; the eleven instances we report are consequently a lower bound. We coded front-office references to the log without a formal inter-rater reliability check, a gap the small instance count made difficult to justify statistically, though both coders’ independent notes agreed on all eleven instances before discussion.

Conclusion

A paper-till, sticker-tag pre-order system built to replace a commercial canteen app processed

6,140 orders over an eight-week pilot with mismatch rates falling from 8.7% to 1.9% as users adjusted, at a modest and largely fixed cost in per-order processing time. An ablation of the order slip's carbon-duplicate stock found no measurable contribution to mismatch rate, isolating the duplicate's value to the audit trail it leaves rather than to transaction accuracy. That audit trail found an unplanned second use within the pilot's first month, as front-office staff began reading a late order timestamp as informal corroboration of a late arrival. Future work might formally evaluate that use's accuracy, something this pilot neither set out to do nor, once it noticed the practice, felt obviously well placed to referee.

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